

KEWEI WANG

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EDUCATION

Courant Institute, New York University

Ph.D. in Mathematics (expected), advised by Georg Stadler

New York, NY

Sept. 2024 - May 2029

Peking University

B.S. in Computational Mathematics

Beijing, China

Sept. 2020 - June 2024

RESEARCH INTERESTS

Optimization under uncertainty; diffusion models; operator learning; Gaussian processes; Bayesian inverse problems.

RESEARCH EXPERIENCE

Hybrid infinite-dimensional SRD for chance-constrained optimization

Dec. 2024 - Present

Graduate research assistant, advised by Georg Stadler, NYU Courant

- Developed hybrid infinite-dimensional spherical-radial decomposition (hiSRD), an unbiased, low-variance, and differentiable estimator of probabilistic constraints that combines subspace SRD with Monte Carlo sampling of the stochastic remainder, eliminating truncation-induced bias while retaining differentiability.
- Quantified the deterioration of finite-dimensional SRD's variance reduction with an $O(K^{-1/2})$ bound as the stochastic dimension K grows; derived explicit hiSRD gradient formulas and proved their almost-sure validity.
- Implemented and validated hiSRD on a PDE-constrained optimal control problem with 4,096 joint state constraints and on chance-constrained Gaussian process kernel optimization using JAX and tinygp, obtaining lower probability-estimation error than standard Monte Carlo together with accurate gradients.

Inverse problems in multipopulation mean field games

Apr. 2023 - Feb. 2024

Undergraduate research assistant, advised by Kui Ren, Columbia University

- Extended inverse-problem theory from single- to multipopulation mean field games, formulating recovery of coupled running and terminal cost functions from initial-time value-function observations.
- Developed a multilinearization framework for the nonlinear, coupled MFG system and proved uniqueness from full multipopulation data; further established recovery of common or state-independent cost functions using measurements from only one population component under mild coupling assumptions.

PUBLICATIONS

Preprints

- [1] **K. Wang** and G. Stadler, "Infinite-dimensional spherical-radial decomposition for probabilistic functions, with application to constrained optimal control and gaussian process regression," *arXiv:2603.19907*, 2026.

Peer-reviewed publications

- [2] K. Ren, N. Soedjak, and **K. Wang**, "Unique determination of cost functions in a multipopulation mean field game model," *Journal of Differential Equations*, vol. 427, pp. 843–867, 2025. [DOI].
- [3] K. Ren, N. Soedjak, **K. Wang**, and H. Zhai, "Reconstructing a state-independent cost function in a mean-field game model," *Inverse Problems*, vol. 40, no. 10, p. 105 010, 2024. [DOI].

POSTER PRESENTATIONS

Infinite-dimensional spherical-radial decomposition for probabilistic functions Apr. 2026

ICERM Workshop: *Simulation-Based Optimization with Applications*

Gaussian Process Regression Subject to Chance Constraints Oct. 2025

IMSI Workshop: *Optimal Control and Decision Making Under Uncertainty for Digital Twins*

TEACHING

Written exam workshop: linear algebra (NYU, Instructor) Summer 2026

Math-UA 132: Math for Economics II (NYU, Teaching assistant) Spring 2026, Fall 2025

AWARDS

- Charles Newman Fellowship 2026
- Henry M. MacCracken Fellowship
Full Ph.D. support at NYU 2024-2029
- Silver Award, Lin Award (Applied and Computational Mathematics) in S.-T. Yau College Student Mathematics Contest June 2023

SKILLS

- **Programming languages:** Python, MATLAB.
- **Tools:** JAX, Firedrake, PyTorch, Git, $\LaTeX$.